

OPTION B INSTITUTE FAMILY

M09 - ETL/ELT + DATA QUALITY + TRANSFORMATION ENGINEERING

Standalone Zero-to-Job-Ready Data Engineer 2026

BEGINNER-FIRST MASTER REPAIR RC4

13 canonical lessons | full 16-stage teacher contract | protected attempt-first
review

This learner book is rebuilt from the frozen canonical technical material. It is not a front-matter patch: every lesson is re-laid out into the shared institute teaching grammar while preserving the course competency, source route, guided work, independent proof and validation boundary.

This book is your teacher

Use the book in order. External videos and official docs are visual companions, not hidden prerequisites. Watch or read only the assigned scope, then close it before the independent task.

The 16-stage lesson loop

Why it matters -> What you learn -> Prerequisite -> Picture it -> New words -> Useful source -> Follow once -> Expected result -> Why it worked -> Beginner mistakes -> Recovery -> Independent task -> Validate -> Teach it back -> Protected review + changed retry -> Evidence/mastery + exact next route.

Attempt-first protection

Never open a protected lesson review before saving a genuine independent attempt. If repair is needed, compare only the weak idea, close the review, solve a changed retry, then validate again. A solution seen during review is repair evidence, not mastery.

What counts as mastery

A reproducible artifact or explanation, an independent validation control, preserved failure/recovery evidence, a changed transfer task completed without the worked answer open, and an own-words explanation. Percent-complete is not a substitute.

How to use your devices

Phone: assigned video/reference, reading, revision. Laptop/PC: every real command, query, file, workbook, code task, validation and evidence artifact. Install only what the current lesson asks for.

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Prerequisite / tool map

Build deterministic ETL/ELT transformations with contracts, reconciliation, tests and incremental behavior.

Item	Rule
Entry prerequisite	Previous module/Gate evidence must be stable.
Primary device	PC/laptop for hands-on work; phone is support only.
External sources	Optional/supporting unless the lesson explicitly marks an official current reference.
Evidence	Module practical + fresh retest + saved evidence; no critical unresolved failure.

Practice / review / Gate route

Lesson teaching -> follow once -> independent attempt -> validate -> save evidence. Only after the genuine attempt may protected review open. If weak: targeted repair -> close review -> changed retry -> revalidate. Finish the module practical/retest route before any downstream Gate that depends on this module.

DE09-01 - ETL vs ELT and transformation boundaries

1. Why it matters

This matters because later ETL/ELT + Data Quality + Transformation Engineering work assumes this competency is reliable, not merely familiar. You must be able to reproduce it and validate the result without copying a finished answer.

2. What you are learning

Choose ETL vs ELT boundaries from trust, compute and ownership rather than slogans. Teach it from zero ETL transforms before final load; ELT lands source-shaped data first and transforms inside the destination. Both need contracts, lineage and reconciliation.

3. Prerequisite / starting state

M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

4. Picture it in your head

VIDEO FIRST route DataTalksClub DE Zoomcamp - Introduction to Analytics Engineering Whole introduction; use it for ETL/ELT and analytics-engineering context. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals.

5. New words before we use them

VIDEO FIRST route DataTalksClub DE Zoomcamp - Introduction to Analytics Engineering Whole introduction; use it for ETL/ELT and analytics-engineering context. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

6. Useful video / source

Source: DataTalksClub DE Zoomcamp — Introduction to Analytics Engineering
Assigned scope: Whole introduction — ETL/ELT boundary and analytics-engineering role
How to use it: Long or short video is allowed when it teaches the competency well. Watch actively without handwritten notes; type/run/follow when relevant. Use the Teaching Book as ready-made notes. Close video/review/AI solving before the independent task.

Note: Book retains our DE-specific boundaries and reliability rules.

[Open source](#)

7. Follow with me once

Draw source -> landing/raw -> validated staging -> intermediate -> serving. Mark where each rule is enforced. Use the supplied RetailPulse files and scripts. Reproduce the demonstrated control once, then close this page and repeat the same reasoning on a slightly changed slice.

8. Expected result / what you should see

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

If you argue only from tool preference, restate the data-risk, recoverability and ownership requirements.

11. If you get stuck

If you argue only from tool preference, restate the data-risk, recoverability and ownership requirements.

12. Small independent task

Given a CSV plus a warehouse, justify which cleanup must happen before load and which business modeling can happen after load.

13. Validate

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

14. Teach it back

In 60-90 seconds explain the rule, one failure mode, one validation and why the chosen boundary protects correctness. Current authoritative reference(s) DataTalksClub 2026 course map MASTER only after a changed task works without the lesson/review open.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE09-02

DE09-02 - Raw, staging, clean/intermediate and serving contracts

1. Why it matters

This matters because later ETL/ELT + Data Quality + Transformation Engineering work assumes this competency is reliable, not merely familiar. You must be able to reproduce it and validate the result without copying a finished answer.

2. What you are learning

Define explicit contracts for raw, staging, intermediate and serving layers. Teach it from zero Raw preserves received evidence. Staging standardizes types/keys while surfacing bad rows. Intermediate expresses reusable business logic. Serving exposes stable consumer grain.

3. Prerequisite / starting state

M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

4. Picture it in your head

VIDEO FIRST route DataTalksClub DE Zoomcamp - dbt Project Structure Project layers and dependency structure; course lab owns the exact layer contracts. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals.

5. New words before we use them

VIDEO FIRST route DataTalksClub DE Zoomcamp - dbt Project Structure Project layers and dependency structure; course lab owns the exact layer contracts. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

6. Useful video / source

Source: DataTalksClub DE Zoomcamp — dbt Project Structure

Assigned scope: Whole focused lesson — project layers/structure; map to raw → staging → intermediate → serving

How to use it: Long or short video is allowed when it teaches the competency well. Watch actively without handwritten notes; type/run/follow when relevant. Use the Teaching Book as ready-made notes. Close video/review/AI solving before the independent task.

Note: Use as visual teacher; do not replace our contract/evidence task.

[Open source](#)

7. Follow with me once

Inspect the RetailPulse SQLite tables and lineage.json; state grain and allowed mutations at each layer. Use the supplied RetailPulse files and scripts. Reproduce the demonstrated control once, then close this page and repeat the same reasoning on a slightly changed slice.

8. Expected result / what you should see

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

If layers blur together, ask: can I reconstruct source evidence; where is invalid data visible; what table is safe for consumers?

11. If you get stuck

If layers blur together, ask: can I reconstruct source evidence; where is invalid data visible; what table is safe for consumers?

12. Small independent task

Write the four layer contracts before writing transformations.

13. Validate

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

14. Teach it back

In 60-90 seconds explain the rule, one failure mode, one validation and why the chosen boundary protects correctness. Current authoritative reference(s) dbt projects dbt sources MASTER only after a changed task works without the lesson/review open.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE09-03

DE09-03 - Deterministic transformations and production SQL style

1. Why it matters

This matters because later ETL/ELT + Data Quality + Transformation Engineering work assumes this competency is reliable, not merely familiar. You must be able to reproduce it and validate the result without copying a finished answer.

2. What you are learning

Write deterministic transformations whose output is a function of declared inputs and rules. Teach it from zero. Deterministic work avoids hidden manual edits, unstable ordering and ambient state. Production SQL should make grain, filters, joins and calculations readable.

3. Prerequisite / starting state

M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

4. Picture it in your head

VIDEO FIRST route DataTalksClub DE Zoomcamp - dbt Models Deterministic model/dependency mental model. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals.

5. New words before we use them

VIDEO FIRST route DataTalksClub DE Zoomcamp - dbt Models Deterministic model/dependency mental model. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

6. Useful video / source

Source: DataTalksClub DE Zoomcamp — dbt Models

Assigned scope: Whole focused lesson — deterministic SQL models and dependencies

How to use it: Long or short video is allowed when it teaches the competency well. Watch actively without handwritten notes; type/run/follow when relevant. Use the Teaching Book as ready-made notes. Close video/review/AI solving before the independent task.

Note: Course book adds production SQL style and rerun correctness.

[Open source](#)

7. Follow with me once

Run the baseline twice from a fresh database and compare serving controls. Use the supplied RetailPulse files and scripts. Reproduce the demonstrated control once, then close this page and repeat the same reasoning on a slightly changed slice.

8. Expected result / what you should see

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

If reruns differ, isolate nondeterminism: time-dependent defaults, unordered tie selection, mutable inputs or hidden state.

11. If you get stuck

If reruns differ, isolate nondeterminism: time-dependent defaults, unordered tie selection, mutable inputs or hidden state.

12. Small independent task

Rewrite one ambiguous transformation as explicit typed/filter logic and add an independent control query.

13. Validate

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

14. Teach it back

In 60-90 seconds explain the rule, one failure mode, one validation and why the chosen boundary protects correctness. Current authoritative reference(s) dbt models MASTER only after a changed task works without the lesson/review open.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE09-04

DE09-04 - Deduplication and key integrity

1. Why it matters

This matters because later ETL/ELT + Data Quality + Transformation Engineering work assumes this competency is reliable, not merely familiar. You must be able to reproduce it and validate the result without copying a finished answer.

2. What you are learning

Separate exact duplicate removal from business-key conflict resolution. Teach it from zero. An exact duplicate repeats the same received row. A repeated business key with changed attributes is not automatically a duplicate; choose a version/merge rule and prove it.

3. Prerequisite / starting state

M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

4. Picture it in your head

VIDEO FIRST route DataTalksClub DE Zoomcamp - dbt Tests Support for uniqueness/not-null/relationship thinking; dedup implementation is course-owned. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals.

5. New words before we use them

VIDEO FIRST route DataTalksClub DE Zoomcamp - dbt Tests Support for uniqueness/not-null/relationship thinking; dedup implementation is course-owned. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

6. Useful video / source

Source: DataTalksClub DE Zoomcamp — dbt Tests

Assigned scope: Focused tests class — uniqueness/not-null/relationship thinking; book adds dedup key-integrity workflow

How to use it: Long or short video is allowed when it teaches the competency well. Watch actively without handwritten notes; type/run/follow when relevant. Use the Teaching Book as ready-made notes. Close video/review/AI solving before the independent task.

Note: Testing video reinforces the key-integrity objective; independent dedup implementation stays course-owned.

[Open source](#)

7. Follow with me once

RetailPulse contains one exact R10 duplicate. The pipeline quarantines it while preserving source evidence. Use the supplied RetailPulse files and scripts. Reproduce the demonstrated control once, then close this page and repeat the same reasoning on a slightly changed slice.

8. Expected result / what you should see

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

Never add DISTINCT to silence unexpected fan-out. Identify the key/grain first.

11. If you get stuck

Never add DISTINCT to silence unexpected fan-out. Identify the key/grain first.

12. Small independent task

Prove raw received rows, exact duplicate count and one-current-row order_id in staging/serving.

13. Validate

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

14. Teach it back

In 60-90 seconds explain the rule, one failure mode, one validation and why the chosen boundary protects correctness. Current authoritative reference(s) dbt data tests MASTER only after a changed task works without the lesson/review open.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE09-05

DE09-05 - Schema checks and data contracts

1. Why it matters

This matters because later ETL/ELT + Data Quality + Transformation Engineering work assumes this competency is reliable, not merely familiar. You must be able to reproduce it and validate the result without copying a finished answer.

2. What you are learning

Enforce a schema/data contract and quarantine incompatible rows without silently coercing them. Teach it from zero A contract includes required fields, types, domain rules and evolution expectations. Compatibility decisions should be explicit.

3. Prerequisite / starting state

M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

4. Picture it in your head

VIDEO FIRST route Confluent - Data Contracts for Long-Term Schema Management Use contract/schema concepts; batch schema enforcement and quarantine are course-owned. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals.

5. New words before we use them

VIDEO FIRST route Confluent - Data Contracts for Long-Term Schema Management Use contract/schema concepts; batch schema enforcement and quarantine are course-owned. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

6. Useful video / source

Source: Confluent — How to use Data Contracts for Long-Term Schema Management
Assigned scope: 0:00–17:00: data-contract purpose, JSON-blob problems, Avro/Protobuf advantages; book/lab adds batch schema checks and quarantine
How to use it: Watch the assigned contract/schema section, then implement the course schema checks and failure handling. The streaming examples are a concept bridge, not a replacement for the batch contract lab.
Note: Explicitly strengthens data-contract depth found in modern DE programs without adding a second full course.

[Open source](#)

7. Follow with me once

R11 has bad date and R12 misses customer_id; both remain visible in quarantine. Use the supplied RetailPulse files and scripts. Reproduce the demonstrated control once, then close this page and repeat the same reasoning on a slightly changed slice.

8. Expected result / what you should see

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

If a cast fails, preserve the raw value and reason; do not convert every error to zero/null.

11. If you get stuck

If a cast fails, preserve the raw value and reason; do not convert every error to zero/null.

12. Small independent task

Add one safe optional-field evolution proposal and one breaking-change example.

13. Validate

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

14. Teach it back

In 60-90 seconds explain the rule, one failure mode, one validation and why the chosen boundary protects correctness. Current authoritative reference(s) Confluent schemas and data contracts MASTER only after a changed task works without the lesson/review open.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE09-06

DE09-06 - Source-to-target reconciliation

1. Why it matters

This matters because later ETL/ELT + Data Quality + Transformation Engineering work assumes this competency is reliable, not merely familiar. You must be able to reproduce it and validate the result without copying a finished answer.

2. What you are learning

Reconcile source, staging and serving with row and value controls. Teach it from zero Counts alone can pass while money is wrong; sums alone can pass while rows are duplicated. Reconciliation needs both plus exclusion rules.

3. Prerequisite / starting state

M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

4. Picture it in your head

VIDEO FIRST route DataTalksClub DE Zoomcamp - dbt Sources Sources/freshness context; reconciliation is proven in the local lab. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals.

5. New words before we use them

VIDEO FIRST route DataTalksClub DE Zoomcamp - dbt Sources Sources/freshness context; reconciliation is proven in the local lab. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

6. Useful video / source

Source: DataTalksClub DE Zoomcamp — dbt Sources

Assigned scope: Sources/freshness context; pair with our source-to-target reconciliation lab

How to use it: Long or short video is allowed when it teaches the competency well. Watch actively without handwritten notes; type/run/follow when relevant. Use the Teaching Book as ready-made notes. Close video/review/Al solving before the independent task.

Note: Reconciliation proof remains our controlled task.

[Open source](#)

7. Follow with me once

Baseline: 14 received, 11 accepted unique, 3 quarantined, 10 completed serving rows, revenue 2990.00. Use the supplied RetailPulse files and scripts. Reproduce the demonstrated control once, then close this page and repeat the same reasoning on a slightly changed slice.

8. Expected result / what you should see

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

If totals disagree, compare layer by layer until the first divergence; repair there.

11. If you get stuck

If totals disagree, compare layer by layer until the first divergence; repair there.

12. Small independent task

Write controls for source rows, accepted rows, quarantined rows, serving rows, serving revenue and Cancelled exclusion.

13. Validate

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

14. Teach it back

In 60-90 seconds explain the rule, one failure mode, one validation and why the chosen boundary protects correctness. Current authoritative reference(s) dbt sources MASTER only after a changed task works without the lesson/review open.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE09-07

DE09-07 - Incremental processing and safe reruns

1. Why it matters

This matters because later ETL/ELT + Data Quality + Transformation Engineering work assumes this competency is reliable, not merely familiar. You must be able to reproduce it and validate the result without copying a finished answer.

2. What you are learning

Process new/changed records incrementally and make reruns safe. Teach it from zero Incremental models need a trustworthy cursor and unique key. Reprocessing the same batch must not create duplicates.

3. Prerequisite / starting state

M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

4. Picture it in your head

VIDEO FIRST route DataTalksClub DE Zoomcamp 2026 - From APIs to Warehouses with dlt Incremental-loading visual; course lab proves safe rerun/state behavior. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals.

5. New words before we use them

VIDEO FIRST route DataTalksClub DE Zoomcamp 2026 - From APIs to Warehouses with dlt Incremental-loading visual; course lab proves safe rerun/state behavior. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

6. Useful video / source

Source: DataTalksClub DE Zoomcamp 2026 — From APIs to Warehouses with dlt

Assigned scope: Long-form main class — incremental loading; book/lab proves safe rerun and watermark/state behavior

How to use it: Long or short video is allowed when it teaches the competency well. Watch actively without handwritten notes; type/run/follow when relevant. Use the Teaching Book as ready-made notes. Close video/review/AI solving before the independent task.

Note: Use the long class once; later revisit only the incremental section.

[Open source](#)

7. Follow with me once

Apply R6 update + R14 new record; serving becomes 11 rows / 3490.00. Rerun selects 0. Use the supplied RetailPulse files and scripts. Reproduce the demonstrated control once, then close this page and repeat the same reasoning on a slightly changed slice.

8. Expected result / what you should see

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

If a changed key duplicates, check unique-key logic and $\geq$ cursor boundary before changing data.

11. If you get stuck

If a changed key duplicates, check unique-key logic and $\geq$ cursor boundary before changing data.

12. Small independent task

Explain why updated_at is the cursor and order_id is the merge key; show rerun evidence.

13. Validate

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

14. Teach it back

In 60-90 seconds explain the rule, one failure mode, one validation and why the chosen boundary protects correctness. Current authoritative reference(s) dbt incremental models dlt incremental loading MASTER only after a changed task works without the lesson/review open.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE09-08

DE09-08 - Failure behavior, retries, quarantine and atomic state

1. Why it matters

This matters because later ETL/ELT + Data Quality + Transformation Engineering work assumes this competency is reliable, not merely familiar. You must be able to reproduce it and validate the result without copying a finished answer.

2. What you are learning

Keep destination and checkpoint/state atomic when a batch fails validation. Teach it from zero State must advance only after the destination commit is durable. Retry transient work; quarantine record-level defects; fail the run for contract-breaking conditions according to policy.

3. Prerequisite / starting state

M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

4. Picture it in your head

VIDEO FIRST route DataTalksClub DE Zoomcamp - dbt Commands Build/command support only; failure atomicity is proven by the course failure injection. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals.

5. New words before we use them

VIDEO FIRST route DataTalksClub DE Zoomcamp - dbt Commands Build/command support only; failure atomicity is proven by the course failure injection. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

6. Useful video / source

Source: DataTalksClub DE Zoomcamp — dbt Commands

Assigned scope: Focused commands/build class; course failure lab adds retry/quarantine/atomic-state behavior

How to use it: Long or short video is allowed when it teaches the competency well. Watch actively without handwritten notes; type/run/follow when relevant. Use the Teaching Book as ready-made notes. Close video/review/AI solving before the independent task.

Note: Video shows operation; our failure scenarios prove reliability.

[Open source](#)

7. Follow with me once

Failure batch contains a numeric defect. `fail_on_quarantine=True` returns `FAILED_VALIDATION` and leaves watermark unchanged. Use the supplied RetailPulse files and scripts. Reproduce the demonstrated control once, then close this page and repeat the same reasoning on a slightly changed slice.

8. Expected result / what you should see

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

If state moved on failure, treat it as a correctness bug, not a logging issue.

11. If you get stuck

If state moved on failure, treat it as a correctness bug, not a logging issue.

12. Small independent task

Inject one failure and prove destination row count and state are unchanged.

13. Validate

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

14. Teach it back

In 60-90 seconds explain the rule, one failure mode, one validation and why the chosen boundary protects correctness. Current authoritative reference(s) dbt commands MASTER only after a changed task works without the lesson/review open.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE09-09

DE09-09 - Lineage, run metadata and auditability

1. Why it matters

This matters because later ETL/ELT + Data Quality + Transformation Engineering work assumes this competency is reliable, not merely familiar. You must be able to reproduce it and validate the result without copying a finished answer.

2. What you are learning

Produce lineage and run metadata sufficient to explain what ran and what changed. Teach it from zero. Auditability links source file, run id, layer, counts, state before/after, status and output controls. Lineage records dependencies and grains.

3. Prerequisite / starting state

M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

4. Picture it in your head

VIDEO FIRST route DataTalksClub DE Zoomcamp - dbt Documentation Documentation/lineage context; run metadata is generated locally. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals.

5. New words before we use them

VIDEO FIRST route DataTalksClub DE Zoomcamp - dbt Documentation Documentation/lineage context; run metadata is generated locally. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

6. Useful video / source

Source: DataTalksClub DE Zoomcamp — dbt Documentation

Assigned scope: Whole focused lesson — documentation/lineage context

How to use it: Long or short video is allowed when it teaches the competency well. Watch actively without handwritten notes; type/run/follow when relevant. Use the Teaching Book as ready-made notes. Close video/review/AI solving before the independent task.

Note: Course adds run metadata and auditability evidence.

[Open source](#)

7. Follow with me once

Inspect run_audit plus lineage.json produced by the pipeline. Use the supplied RetailPulse files and scripts. Reproduce the demonstrated control once, then close this page and repeat the same reasoning on a slightly changed slice.

8. Expected result / what you should see

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

If you cannot answer which input/run produced a row, the handoff is incomplete.

11. If you get stuck

If you cannot answer which input/run produced a row, the handoff is incomplete.

12. Small independent task

Trace R6 from source update through staging to serving and identify its run metadata.

13. Validate

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

14. Teach it back

In 60-90 seconds explain the rule, one failure mode, one validation and why the chosen boundary protects correctness. Current authoritative reference(s) dbt documentation MASTER only after a changed task works without the lesson/review open.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE09-10

DE09-10 - Build and defend a reliable batch pipeline

1. Why it matters

This matters because later ETL/ELT + Data Quality + Transformation Engineering work assumes this competency is reliable, not merely familiar. You must be able to reproduce it and validate the result without copying a finished answer.

2. What you are learning

Build and defend a reliable batch pipeline end to end. Teach it from zero A reliable batch path is not just transformation code: contracts -> validation -> quarantine -> deterministic transform -> reconciliation -> tests -> atomic state -> metadata -> rerun proof.

3. Prerequisite / starting state

M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

4. Picture it in your head

VIDEO FIRST route DataTalksClub DE Zoomcamp - dbt Project Structure Review structure -> sources -> models -> tests -> docs, then close the video and build. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals.

5. New words before we use them

VIDEO FIRST route DataTalksClub DE Zoomcamp - dbt Project Structure Review structure -> sources -> models -> tests -> docs, then close the video and build. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

6. Useful video / source

Source: DataTalksClub DE Zoomcamp — dbt video course sequence
Assigned scope: Main-class review: project structure → sources → models → tests → docs; then close video and build independently
How to use it: Long or short video is allowed when it teaches the competency well. Watch actively without handwritten notes; type/run/follow when relevant. Use the Teaching Book as ready-made notes. Close video/review/AI solving before the independent task.
Note: Do not copy a finished project. This is pre-build consolidation only.
[Open source](#)

7. Follow with me once

Run m09_pipeline.py from a clean output folder and read M09_SMOKE_RESULT.json. Use the supplied RetailPulse files and scripts. Reproduce the demonstrated control once, then close this page and repeat the same reasoning on a slightly changed slice.

8. Expected result / what you should see

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

When a result is plausible but controls fail, trust controls and investigate.

11. If you get stuck

When a result is plausible but controls fail, trust controls and investigate.

12. Small independent task

Rebuild the pipeline on a changed file and produce an evidence packet, not only a success message.

13. Validate

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

14. Teach it back

In 60-90 seconds explain the rule, one failure mode, one validation and why the chosen boundary protects correctness. Current authoritative reference(s) DataTalksClub analytics engineering module MASTER only after a changed task works without the lesson/review open.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE09-11

DE09-11 - dbt-style project anatomy, sources and dependency graph

1. Why it matters

This matters because later ETL/ELT + Data Quality + Transformation Engineering work assumes this competency is reliable, not merely familiar. You must be able to reproduce it and validate the result without copying a finished answer.

2. What you are learning

Read a dbt-style project as a dependency graph instead of disconnected SQL files. Teach it from zero source() represents declared external/source relations; ref() creates model dependencies; project/YAML config adds metadata/tests/materialization behavior.

3. Prerequisite / starting state

M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

4. Picture it in your head

VIDEO FIRST route DataTalksClub DE Zoomcamp - dbt Project Structure Whole focused project-anatomy lesson. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals.

5. New words before we use them

VIDEO FIRST route DataTalksClub DE Zoomcamp - dbt Project Structure Whole focused project-anatomy lesson. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

6. Useful video / source

Source: DataTalksClub Zoomcamp — dbt project structure

Assigned scope: Whole focused lesson

How to use it: Watch the assigned visual first, then use the Teaching Book for the course-specific build. Close the visual before the independent task.

[Open source](#)

7. Follow with me once

Open dbt_style_project: source -> stg_orders -> int_completed_orders -> fct_sales. Use the supplied RetailPulse files and scripts. Reproduce the demonstrated control once, then close this page and repeat the same reasoning on a slightly changed slice.

8. Expected result / what you should see

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

If the graph is unclear, follow source()/ref() edges before reading every SQL expression.

11. If you get stuck

If the graph is unclear, follow source()/ref() edges before reading every SQL expression.

12. Small independent task

Sketch the DAG, identify each model grain and explain why direct raw -> mart logic is harder to govern.

13. Validate

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

14. Teach it back

In 60-90 seconds explain the rule, one failure mode, one validation and why the chosen boundary protects correctness. Current authoritative reference(s) dbt projects dbt sources MASTER only after a changed task works without the lesson/review open.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE09-12

DE09-12 - Executable data tests, quality severity and CI boundaries

1. Why it matters

This matters because later ETL/ELT + Data Quality + Transformation Engineering work assumes this competency is reliable, not merely familiar. You must be able to reproduce it and validate the result without copying a finished answer.

2. What you are learning

Turn quality expectations into executable tests with severity and CI boundaries. Teach it from zero. Blocking tests protect invariants such as required keys/uniqueness. Warning tests surface risk that may not justify stopping every run. CI should fail only on deliberately blocking conditions.

3. Prerequisite / starting state

M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

4. Picture it in your head

VIDEO FIRST route DataTalksClub DE Zoomcamp - dbt Tests Whole focused tests lesson; course lab adds severity/CI boundaries. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals.

5. New words before we use them

VIDEO FIRST route DataTalksClub DE Zoomcamp - dbt Tests Whole focused tests lesson; course lab adds severity/CI boundaries. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

6. Useful video / source

Source: DataTalksClub Zoomcamp — dbt tests

Assigned scope: Whole focused lesson

How to use it: Watch the assigned visual first, then use the Teaching Book for the course-specific build. Close the visual before the independent task.

[Open source](#)

7. Follow with me once

The SQLite smoke runs unique key, required customer, positive revenue and reconciliation tests. Use the supplied RetailPulse files and scripts. Reproduce the demonstrated control once, then close this page and repeat the same reasoning on a slightly changed slice.

8. Expected result / what you should see

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

Do not make every anomaly fatal and do not make critical integrity checks warnings.

11. If you get stuck

Do not make every anomaly fatal and do not make critical integrity checks warnings.

12. Small independent task

Classify five proposed checks as BLOCK or WARN and defend the business consequence.

13. Validate

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

14. Teach it back

In 60-90 seconds explain the rule, one failure mode, one validation and why the chosen boundary protects correctness. Current authoritative reference(s) dbt data tests MASTER only after a changed task works without the lesson/review open.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: DE09-13

DE09-13 - Incremental models, materialization, documentation and handoff

1. Why it matters

This matters because later ETL/ELT + Data Quality + Transformation Engineering work assumes this competency is reliable, not merely familiar. You must be able to reproduce it and validate the result without copying a finished answer.

2. What you are learning

Explain materialization/incremental behavior and hand off a reproducible project. Teach it from zero Views, tables and incremental models trade freshness, compute and persistence. Current dbt incremental models require valid SQL for both full/incremental paths; unique_key supports updates instead of blind append.

3. Prerequisite / starting state

M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

4. Picture it in your head

VIDEO FIRST route DataTalksClub DE Zoomcamp - dbt Models Models/materialization support; pair with current dbt incremental-model documentation. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals.

5. New words before we use them

VIDEO FIRST route DataTalksClub DE Zoomcamp - dbt Models Models/materialization support; pair with current dbt incremental-model documentation. Starting state / prerequisite M08 ingestion mastery plus SQL/Python fundamentals. Keep the protected review closed until a genuine attempt is saved.

6. Useful video / source

Source: DataTalksClub Zoomcamp — dbt models + commands

Assigned scope: Whole focused lesson; then use dbt commands lesson if needed

How to use it: Watch the assigned visual first, then use the Teaching Book for the course-specific build. Close the visual before the independent task.

Note: Companion command video: <https://www.youtube.com/watch?v=t4OeWHW3SsA>

[Open source](#)

7. Follow with me once

Inspect fct_sales.sql with materialized=incremental, unique_key=order_id and on_schema_change=fail; compare it with the executed SQLite equivalent. Use the supplied RetailPulse files and scripts. Reproduce the demonstrated control once, then close this page and repeat the same reasoning on a slightly changed slice.

8. Expected result / what you should see

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

9. Why it worked

The worked path succeeds because it begins from the stated starting state, applies the lesson rule deliberately, and checks an observable result before the independent transfer. The validation—not the fact that a command or click completed—is what supports the claim.

10. Common beginner mistakes

If incremental logic changes historical meaning, plan a controlled rebuild/backfill rather than assuming future runs repair history.

11. If you get stuck

If incremental logic changes historical meaning, plan a controlled rebuild/backfill rather than assuming future runs repair history.

12. Small independent task

Write a handoff: inputs, grain, model DAG, tests, state/cursor, full-refresh/backfill decision, known limitations and rerun commands.

13. Validate

Save source/layer grain, row counts, key checks, at least one value reconciliation, run/state evidence where relevant, and one failure-mode note. A success message alone is not evidence.

14. Teach it back

In 60-90 seconds explain the rule, one failure mode, one validation and why the chosen boundary protects correctness. Current authoritative reference(s) dbt incremental models dbt materializations dbt documentation MASTER only after a changed task works without the lesson/review open.

15. Protected review + changed retry

Save the genuine independent attempt first. Only then may the matching protected review open. Diagnose the exact weak competency, close the review, and complete a changed retry without the answer visible. The corrected-with-review attempt is repair evidence, not mastery.

16. Evidence / mastery + exact next route

Complete the matching guided/independent work with review material closed; save the actual result/error and validation evidence. Exact next route: Proceed to the module practical/Gate route after the module mastery checklist.

Module mastery checklist + revision quick reference

- ☐ Every lesson has a saved genuine independent attempt.
- ☐ Every lesson has at least one independent validation/control appropriate to the tool.
- ☐ Protected reviews were opened only after attempts; any weak lesson has a changed retry after review was closed.
- ☐ You can teach back each lesson in your own words and name one realistic failure mode.
- ☐ Required module practical/retest evidence is saved and no critical defect is unresolved.
- ☐ D+1, D+4, D+10 and D+30 revision events are scheduled for weak or high-value competencies.

Revision cadence

D+1: fresh micro-task. D+4: mixed retrieval. D+10: changed/failure case. D+30: interview/project retrieval. Always attempt from memory before reopening a source.

Release boundary

This book is a static learner artifact. It does not prove live tool/cloud execution or a protected Gate pass. Runtime evidence must be recorded separately when the curriculum requires it.